

# The influence of human behavioral characteristics on soundscape perception in urban parks: Subjective and observational approaches

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## ABSTRACT

With an emphasis on the role of parks in urban environments, studies on soundscape, which assess sound quality, have focused on human perception. In particular, various studies have been conducted on the relationship between human behavioral characteristics and their sound environment. However, there is still insufficient research on human behavioral characteristics in the context of urban parks. The purpose of this study was to investigate the effect of human behavioral characteristics on perceptions of soundscapes in urban parks. Participants were asked to subjectively evaluate the soundscapes of three parks in Paris, France, using three cognitive stages: indicators (perceived audio and visual elements), emotions (semantic expression), and cognition (overall quality of the park's soundscape). The human behavioral characteristics that the participants witnessed were categorized into whether there was social interaction, whether people remained in the park or were passing through, and what kind of activities took place. To secure high ecological validity, audio-visual information was provided to the participants using virtual reality equipment. The results demonstrated that sounds made by other people decreased the perceived tranquility or peacefulness while the experience of the park as being dynamic and lively increased. We also found that the primary determinant of park preferences was not the type of human activities but the presence or absence of people.

## 1. Introduction

### 1.1. Soundscape research in urban parks

As the negative effects of noise in urban environments on people's health and social well-being became apparent (Passchier-Vermeer & Passchier, 2000; Pulles, Biesiot, & Stewart, 1990; Tzoulas et al., 2007), parks have come to be regarded as essential elements of sustainable and healthy cities. Subsequently, the positive effects of urban parks on the ecological environment and public good have been confirmed (Brown, Schebella, & Weber, 2014; Tse et al., 2012). In particular, urban parks are believed to offer important social and psychological advantages (sources of positive emotions and beneficial services), and thereby enrich human lives (Chiesura, 2004). As a result, visual aesthetics and sound quality have become important factors in designing urban landscapes (Zhang, Yang, & Yu, 2006). In this context, the concept of soundscapes, focusing on people's perception of sound, has emerged (Corburn, 2009; Kang & Schulte-Fortkamp, 2018).

International standard ISO 12913-1 (2014) defines soundscapes as "acoustic environment as perceived or experienced and/or understood by a person or people, in context." To evaluate soundscape perception

in various spaces such as parks and open spaces, many studies have examined the relationships between indicators (e.g., acoustical indices and contextual factors) and descriptors (e.g., comfort, tranquility, and soundscape quality; Brown, Kang, & Gjestland, 2011; Kang & Zhang, 2010; Yu & Kang, 2009). Among the studies on soundscape in urban parks, there have been attempts to identify the factors that positively or negatively affect perception using interviews (Szeremeta & Zannin, 2009); categorize parks according to their physical sound (Brambilla, Gallo, & Zambon, 2013); explore non-sonic components (presence of greenery, tranquility, and so on) that positively affect soundscape besides sonic components, such as sound pressure level (Brambilla, Gallo, Asdrubali, & D'Alessandro, 2013); and investigate the components that affect soundscape comfort through audio-visual interactions (Zhang, Zhao, Zenz, & Qiu, 2019). Deep learning methods to classify sounds made in parks using an auditory attention model have also been used (Filipan, Boes, Oldoni, De Coensel, & Botteldooren, 2014).

Based on previous studies, a soundscape perception model for urban planning and design has been proposed (Herranz-Pascual, Aspuru, & García, 2010; Hong & Jeon, 2015; Liu, Kang, Luo, & Behm, 2013). Despite these efforts, urban soundscapes are occasionally unsatisfactory because there is a gap between academia and practice that prevents

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research findings from being utilized in the decision-making process, and additional interdisciplinary research on soundscapes is necessary (Aletta & Xiao, 2018; Meng & Kang, 2016). In this regard, Aletta and Xiao (2018) surveyed eleven early-career soundscape researchers and practitioners and found that there is a lack of research on the relationships between human behavioral characteristics and soundscape perception.

### 1.2. The relationship between human behavioral characteristics and the sound environment

Recent studies on the relationships between soundscape and human behavioral characteristics in the urban landscape have been based on the effect of the sound environment on human behavior and the effect of human behavioral characteristics on acoustic cognition. More research has focused on the former, focusing on how the sound environment changes human behavioral characteristics (Aletta, Lepore, Kostara-Konstantinou, Kang, & Astolfi, 2016; Bild, Pfeffer, Coler, Bertolini, & Rubin, 2018; Davies et al., 2013; Meng, Zhao, & Kang, 2018; Witchel, Lavia, Westling, Healy, Needham, & Chockalingam, 2013). First, Davies et al. (2013) found that natural sound in urban spaces could have a positive effect on human behavior. Witchel et al. (2013) proposed a method to improve crowd behavior and decrease anti-social behavior by changing the sound and music of the main street of a city. Additionally, Aletta et al. (2016) found that music has a clear effect on the duration of stay in open public spaces (shopping streets), and the duration of stay is longer when classical music is provided compared to when there is no music. In a subsequent study, Meng et al. (2018) specifically examined the effects of typical music (a pop song called “Free to Fly”) on the movement and non-movement behavior of people in a park. They found that the music did not affect walking speed, but that the walking route was more centralized in leisure squares. Furthermore, Bild et al. (2018) examined the effect of people’s activities on their soundscape evaluation in public spaces. They concluded that solitary and socially interactive respondents evaluate soundscapes differently, depending on their activities. As such, studies examining the effects of the sound environment on human behavioral characteristics have been conducted actively in open public spaces.

In soundscape studies, various proposals have been developed to quantify human behavioral characteristics in urban microscale units: Guastavino (2007) attempted to distinguish between different types of activity (e.g., taking a walk or having a drink), and Wang (2014) classified human behavioral characteristics into movement and non-movement through observational methods. Jia (2012) used a one-two-multiple people classification for the number of people involved. Subsequent studies have attempted to further refine activities into sitting, standing, and loitering (Lepore, Aletta, Astolfi, & Kang, 2016). In a recent study, Aletta et al. (2016) used Behavioral Observation Research Interactive Software (BORIS project; Friard & Gamba, 2016) to quantify human behavioral characteristics by classifying activities (chatting, eating-drinking, loitering, smoking, talking on the phone), groups (alone/group), and postures (sitting/standing). As such, human behavioral characteristics can be classified using various methods and criteria based on the research purpose and the experimental design.

Little research has been conducted on the effects of human behavioral characteristics on soundscape perception. However, the initial research has focused on external spaces, such as shopping streets (Meng & Kang, 2016, 2013; Meng, Sun, & Kang, 2017). These studies found that even with the same shopping distance, the subjective loudness increased as the crowd density increased, and thus the acoustic comfort changed. Moreover, the evaluation of acoustic comfort varied according to sound-related and music-related activities in open space. In general, the sound generated by people has the potential to generate strong attention due to the degree of implication of the perceiver (Viollon, Lavandier, & Drake, 2002). In other words, the influence of other people on the audio-visual perception of space is very important.

Guastavino (2006) concluded that the most relevant component in urban soundscape is human presence and activity. Multidisciplinary research by Davies et al. (2013) also stated that humans derive the meaning of soundscapes from human activity and behavior.

Additionally, the soundscape affordance varies depending on human activity and the appropriateness of the spatial sound environment. Subsequently, it is necessary to examine the effects of human behavioral characteristics on soundscape perception (Nielbo, Steele, & Guastavino, 2013). Although urban parks are places where people are active, no research has investigated the effects of human behavioral characteristics on urban park soundscape perception, and this research fills this gap.

### 1.3. Audio-visual approaches for soundscape assessment

The urban environment is an ecological entity that comprises soundscapes, landscapes, and people (Jeon & Jo, 2020; Jo & Jeon, 2020). Therefore, in interpreting urban environments, the various human senses and the physical environment should be considered together (Herranz-Pascual et al., 2010). Many studies have examined the soundscape and landscape relationship from an audio-visual point of view (Joynt & Kang, 2010; Liu, Kang, Behm, & Luo, 2014; Pheasant, Horoshenkov, Watts, & Barrett, 2008; Southworth, 1969). First, in terms of soundscape perception, the research demonstrated that visual stimuli, such as landscapes, influence the comfort/preference assessment of soundscapes (Hong & Jeon, 2013; Preis, Kociński, Hafke-Dys, & Wrzosek, 2015). Additionally, a larger-scale approach demonstrated that landscape spatial patterns from a macroscopic perspective have a more significant impact than the on-site landscape composition seen from a microscopic perspective (Liu et al., 2014). Conversely, in terms of landscape cognition, Carles, Barrio, and de Lucio (1999) demonstrated that the sound of space and the congruence or coherence of the landscape influences a subject’s preferences. Studies on the effects of various sounds on the natural landscape have demonstrated that the sound of water had a more positive effect than other natural sounds (e.g., bird calls and a breeze through the foliage; Benfield, Bell, Troup, & Soderstrom, 2010; Jeon, Lee, You, & Kang, 2010). A recent comprehensive study (Ren, Kang, Zhu, & Wang, 2018) conducted on the relationship between soundscapes and landscapes concluded that the composition of soundscapes affected the preference of landscape elements (vegetation, mountains, sky, or no vegetation). Therefore, soundscape perception is closely related to the composition of the landscape; consequently, they need to be interpreted together from an audio-visual point of view.

To assess these multi-sensory soundscapes in a controlled laboratory environment, a realistic and immersive audio-visual reproduction system with a high ecological validity is necessary. Recently, several studies have attempted to evaluate soundscapes using virtual reality (VR) technology (Liu & Kang, 2018; Sanchez, Van Renterghem, Sun, De Coensel, & Botteldooren, 2017; Sun et al., 2019; Viollon et al., 2002). Furthermore, VR technology has been used to evaluate the sound environment on external space (Maffei, Masullo, Pascale, Ruggiero, & Puyana-Romero, 2015; Puyana-Romero, Lopez-Segura, Maffei, Hernández-Molina, & Masullo, 2017). These comparative studies evaluated the global, acoustic, and visual quality of a real environment and the VR reproduced environment for the same urban space and found that there is no difference between VR and the real environment. Additionally, Hong et al. (2019) concluded that the first-order ambisonic reproduction method, which is based on headphone-based head rotation, is realistic for overall soundscape quality, and sound-related and environment-related spatial attributes. Therefore, although subjective evaluation methods using VR technology cannot provide the same experience, in reality, they can be used to evaluate soundscapes while providing an immersive experience in a limited laboratory environment. In addition, VR has the advantage that it can provide the same stimulation conditions to all participants. Therefore, an immersive VR

environment was used in this study.

#### 1.4. Research objectives

This study aimed to examine the audio-visual aspects of the effect of human behavioral characteristics on soundscape perception changes in urban parks. From the discussion above, the following research questions were developed: 1) What are the effects of human behavioral characteristics on the perception of soundscapes in urban parks from an audio-visual perspective? 2) How does the temporal variation of human activities in urban parks correlate with changes in soundscape perception?

## 2. Methods

### 2.1. Study design

This study collected audio-visual soundscape stimuli from three urban parks and analyzed the changes in soundscape perception caused by changes in human behavioral characteristics using the type of park and time as variables. Three different types of parks were selected (four evaluation sites in each), and five temporal variations were set 30 s apart from one another for each evaluation site. The study used the two within-subjects design on the same participant group. To assess the same experimental stimuli, a VR-based laboratory experiment, and a questionnaire composed of three parts (indicators, emotions, and cognition) were used to quantify soundscape perception. Human behavioral characteristics were quantified and grouped into different categories (social interaction, stay conditions, and activity type) through observations.

### 2.2. Soundscape stimulus

#### 2.2.1. Case study area

To examine the behavioral characteristics of people in urban parks, and their impact on soundscape perception, three parks with different uses that are located near the Seine River in Paris, France, were selected. Each park has a different size and use. Arènes de Lutèce (Park A) is the smallest park and is surrounded by dwellings. Compared to the other parks, it is relatively calm; a few people spend some time there, and it has a courtyard in the center. Champ de Mars (Park C) is a large park of 245,000 m<sup>2</sup>, and it includes the Eiffel Tower as a Paris landmark and tourist destination. Jardin des Tuileries (Park J) is 280,000 m<sup>2</sup> and is similar to Park C as it is a tourist destination, but it is also used as a place to relax. Therefore, Park J is a combination of Park A's and Park C's functionalities and population characteristics. Twelve representative points were selected by choosing four points on each park's walking path. The equirectangular panoramic views taken at each point are illustrated in Fig. 1, and the physical characteristics of the evaluation points are illustrated in Table 1.

#### 2.2.2. Stimulus generation for the VR environment

The audio-visual recording and reproduction methods for the VR application are detailed in Appendix I.

### 2.3. Assessment of soundscape perception

#### 2.3.1. Participants

A total of 30 participants (male = 17, female = 13) aged from 20 to 31 years (mean age: 25.2, standard deviation: 2.11) took part in the soundscape evaluation. The sample size was calculated using G-power software version 3.0, and 27 participants were required for the medium effect size ( $f = 0.25$ ) with a statistical power of 0.8. Based on the results, the final number of participants was set to 30.

To reduce the variation, students from the same university enrolled in architecture, urban, and sound-related studies were selected. To

reduce observation bias, they were trained in the soundscape evaluation method through an audio-visual example stimulation recorded in the city center (Hong & Jeon, 2017). However, the final purpose of this experiment was not divulged to avoid prejudice in the evaluation response. Additionally, they were thoroughly familiar with the use of the experimental equipment, signed a consent form indicating their voluntary participation, and were paid a participation fee afterward. The participants had normal hearing.

#### 2.3.2. Questionnaire

The questionnaire for assessing soundscape perception was composed of three parts (indicators, emotions, and cognition) according to the perception process, as illustrated in Table 2. Emotions and cognition interact in complex ways, but emotions are related to "the conscious (or unconscious) evaluation of events" and can be defined as "states elicited by rewards and punishers." Cognition can be defined as comprehensively controlled processes such as "memory, attention, problem-solving, and planning" and "acquiring knowledge and understanding through thought, experience, and the senses" (Pessoa, 2008). In other words, emotions represent the affective state and affect cognition, which is a comprehensive judgment. From this perspective, a questionnaire was constructed by separating emotions and cognition.

First, to examine the identification of the audio and visual indicators that constitute the sound environment at each evaluation point, the sound was evaluated with four elements (traffic, humans, birdsong, and water) and the visuals with eight elements (vehicles, buildings, roads, people, open space, greenery, water, and sky) using a five-point Likert scale. The type of sound was set by referring to the five classification systems of previous soundscape assessment studies (Hong & Jeon, 2015; Liu et al., 2013) and the Soundscape Questionnaire Protocol (traffic, fan, other noise, sounds from people, natural sounds; Berglund & Nilsson, 2006; Nilsson & Berglund, 2006). For visual elements, the indicators (vehicles, buildings, roads, open space, greenery, water, people, and sky) used to evaluate the visual environment in landscape studies were selected (Hong & Jeon, 2013; Jiang, Masullo, Maffei, Meng, & Vorländer, 2018; Liu et al., 2014).

Second, in the emotions section, to examine the affective quality of the soundscape, four studies (Axelsson, Nilsson, & Berglund, 2010; Kang & Zhang, 2010; Kawai, Kojima, Hirate, & Yasuoka, 2004; Raimbault, Lavandier, & Bérengier, 2003) were used to develop a five-point Likert scale for eight adjective attributes (pleasant, unpleasant, eventful, uneventful, exciting, monotonous, calm, and chaotic).

Finally, in the cognition section, the participants' overall perceptions of urban parks were defined as perceived loudness, perceived crowd density, and comfort, and evaluated on a five-point Likert scale. The various direct outcomes and indirect outcomes presented for the soundscape preference assessment by Brown et al. (2011) and the descriptors that were used to examine the preferences for commercial space behavior by Meng et al. (2017) were used.

#### 2.3.3. Procedures

All subjects were provided with the same stimuli under the same laboratory conditions. A total of 12 stimuli consisted of 150 s at each evaluation point, and each stimulus consisted of 60 stimuli (12 locations  $\times$  5 times) separated by 30 s, to observe the temporal variation over time (Aletta & Kang, 2018). As illustrated in Table 1, the regeneration sequence was provided twice in the order of A1 to J4, and the participants could play it as many times as they wanted. There was a 10-second silence interval between each assessment stimulus to minimize the memory impact of the previous stimulus (Baddeley, 1997). The experiment was conducted in a semi-anechoic chamber, with a very low background noise of about 30 dBA. Open type headphones (Sennheiser HD-650) were used to provide sound information, and VIVE Pro 2 (HMD) was used to provide the visual information. To alleviate the physical discomfort of the participants from wearing the HMD for a long time, the time of each evaluation environment was





Fig. 1. Equirectangular panoramic stills from the spherical videos of selected evaluation position for the experiment (Source: Google Earth).

**Table 1**  
Acoustic Parameters and Details of the Selected Locations [dB].

| Location | Description                                            | $L_{Aeq}$ | $L_{A10-A90}$ | $L_{Ceq-Aeq}$ |
|----------|--------------------------------------------------------|-----------|---------------|---------------|
| A1       | Green area with benches, near the courtyard            | 54.5      | 12.4          | 21.8          |
| A2       | Green area with lots of trees, with public seating     | 42.5      | 5.0           | 28.4          |
| A3       | Near the park entrance, close to the road, many trees  | 54.8      | 22.1          | 22.4          |
| A4       | Open square, center of the courtyard                   | 60.4      | 23.9          | 20.7          |
| C1       | Pedestrianized square, sand-bed                        | 67.6      | 21.9          | 17.8          |
| C2       | Large green lawn, away from traffic                    | 72.1      | 22.4          | 16.1          |
| C3       | Near a busy road, near a small fountain                | 69.6      | 19.7          | 17.5          |
| C4       | Near the Eiffel Tower, many people                     | 67.9      | 22.1          | 18.4          |
| J1       | Near a big pond, with public seating, small green area | 65.0      | 24.4          | 18.5          |
| J2       | Big fountain, with public seating, large green area    | 64.2      | 18.9          | 19.3          |
| J3       | Near the Arc de Triomphe, open space, many trees       | 56.1      | 12.1          | 21.9          |
| J4       | Near a road, good paved area, near the Louvre Museum   | 66.0      | 17.5          | 19.6          |

limited to 30 min, and sufficient rest time was provided upon the participant's request. We also developed a questionnaire graphic user interface in the VR environment for selecting a questionnaire response using a VIVE controller.

#### 2.4. Observation of human behavioral characteristics

The behavioral analysis of the 60 videos used in the subjective evaluation was conducted to investigate the behavioral characteristics

of the people in each park. It was calculated by computing the number of park users in the 360-degree image implemented as a panoramic view. Based on previous studies (Aletta et al., 2016; Duan, Wagner, Zhang, Wulff, & Brehm, 2018; Palliwoda, Kowarik, & von der Lippe, 2017; Wolf & Wohlfart, 2014), the behavior was categorized into social interaction, stay conditions, and activity type. First, to determine the degree of social interaction, people were classified into whether they were alone or in a group. To investigate the proportion of people remaining in the park (stay conditions), they were also divided into passers (people who pass through the park) and stoppers (people who stay and do activities). The types of activities were divided into five categories: stopping, loitering, walking, jogging or running, and cycling. Although we also analyzed smoking, strolling with a dog, taking a picture, strolling, sweeping (cleaning), and eating (or drinking), the average number of observations was less than five, indicating small amounts of activity within the evaluation point, so we excluded them from the analysis.

#### 2.5. Data analysis

The following statistical analyses were performed using SPSS version 25. First, normality and equal variance tests were conducted. The normality was evaluated through a Shapiro-Wilk's test, and the uniformity test confirmed that the parameter statistics were applicable using the Levene test. A two-way analysis of variance test (ANOVA) was performed to examine the differences in indicators, emotions, and cognition results based on the park type and time changes. Soundscape adjective attributes related to emotions were extracted from the dominant factor using a principal component analysis (PCA). A varimax



**Table 2**  
Questionnaire Details.

| Variable  | Question/statement                                                                                                                                                                                                     | Type of response<br>(5-Likert scale)                                                                                             |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Indicator | To what extent do you currently hear the following types of sound?<br><i>traffic, human, birdsong, water</i>                                                                                                           | –2: Not heard at all<br>–1: Heard a little<br>0: Heard moderately<br>1: Heard a lot<br>2: Dominated completely                   |
|           | To what extent do you currently see the following types of visual elements?<br><i>vehicle, building, road, people, open space, greenery, water, sky</i>                                                                | –2: Not seen at all<br>–1: Seen a little<br>0: Seen moderately<br>1: Seen a lot<br>2: Dominated completely                       |
| Emotion   | To what extent do you agree with the eight attributes below on how you experience the present surrounding sound environment?<br><i>pleasant, unpleasant, eventful, uneventful, exciting, monotonous, calm, chaotic</i> | –2: Strongly disagree<br>–1: Slightly disagree<br>0: Neither disagree nor agree<br>1: Slightly agree<br>2: Strongly agree        |
| Cognition | To what extent do you recognize the loudness in the present surrounding sound environment?                                                                                                                             | –2: Very soft<br>–1: Soft<br>0: Neither soft nor loud<br>1: Loud<br>2: Very loud                                                 |
|           | To what extent do you recognize the crowd density in the present surrounding sound environment?                                                                                                                        | –2: Very uncrowded<br>–1: Uncrowded<br>0: Neither uncrowded nor crowded<br>1: Crowded<br>2: Very crowded                         |
|           | To what extent do you feel comfortable in the present surrounding sound environment?                                                                                                                                   | –2: Very uncomfortable<br>–1: Uncomfortable<br>0: Neither uncomfortable nor comfortable<br>1: Comfortable<br>2: Very comfortable |
|           |                                                                                                                                                                                                                        |                                                                                                                                  |

rotation was applied to determine the orthogonal components, and the eigenvalue was set above one to determine the number of components. To investigate the effects of human behavioral characteristics on emotions and cognition, a Pearson correlation was used. Additionally, a multiple linear regression (MLR) analysis was conducted to investigate the contribution of human behavioral characteristics.

### 3. Results

#### 3.1. Soundscape perception in urban parks

##### 3.1.1. Indicators—perceived audio and visual elements

Fig. 2 illustrates the identification results for the audio and visual indicators. The response percentages refer to the percentage of participants who rated an element above “0: heard/seen moderately.” The results for the dominant sound source were as follows. Park A had 0.91 and 0.94 higher mean responses for birdsong than the other two parks (C and J, respectively). In Park C, the dominant perception of human sound was 33%, which was higher than that of Park A. In C3, up to 77% of the participants recognized human sound as dominant. Table 3 illustrates the two-way ANOVA results for the identification differences of the indicators based on the park type and temporal variation of each park. Traffic sound had a significant interaction between park type and time variation; subsequently, to characterize these interactions, the simple effect of each variable was analyzed. Additionally, the results demonstrated that the identification differences between the parks were significant, and the identification difference with time was significant in Park C and Park J.

Although the influence of temporal variation on sound

identification by park type was not significant, an ANOVA analysis of sound identification for the period at individual evaluation points demonstrated statistically significant differences ( $p < 0.05$ ) for traffic (A2, A3, C1, and C2), human sounds (A1, A3, C3, C4, and J4), and water (J1 and J3). An interesting fact was that the elements that demonstrate cognitive differences over time were mainly sounds that come from objects that move or relocate. Notably, the changes in the identification of human sounds occurred most frequently. This might be explained by the fact that participants reacted sensitively to human sound changes in the park.

The results for the dominant visual elements were as follows. As parks have very different characteristics in terms of their functionality, landscape, and building layout, there were statistically significant differences between parks in the perception of all visual indicators, but in different ways (see Table 3). Participants in Park A had the highest awareness of greenery (54%), and sky (40%) and open space (36%) were relatively higher than other visual indicators due to the open space in the center of the park. The awareness of participants in Park C of people, open space, and the sky was very high at 68%, and because of the surrounding vegetation, awareness of greenery was also high at 44%. The participants in Park J had a higher visual identification of people and sky than participants in Park C, at more than 70%, followed by open space (55%) and greenery (40%). From the one-way-ANOVA on time perception of periods, a statistically significant difference was determined ( $p < 0.05$ ) for vehicles (A3, A4, C3, and C4), people (C1, J2, and J3), and water (C3, J2, and J3). Like the sound elements, they were the elements that moved, especially in the presence of people.

##### 3.1.2. Emotions—perceived affective quality

A PCA was performed on the evaluation results for the soundscapes, and the results are illustrated in Table 4. The Kaiser-Meyer-Olkin measure of sampling adequacy was calculated at 0.71, and Bartlett's test of sphericity was evident in all of the evaluation settings ( $p < 0.01$ ). Soundscape emotions for urban parks were composed of the following three components. Component 1 had an explanatory power of 26.30% and can be expressed as tranquil/peaceful, including the words “uneventful,” “monotonous,” and “calm.” Component 2 had a 24.00% explanatory power and can be interpreted as dynamic/lively, including “pleasant,” “eventful,” and “exciting.” Finally, Component 3 contained “unpleasant” and “chaotic” and can be interpreted as confusing/unpleasant, with an explanatory power of 18.48%.

To understand the emotions of different park soundscapes, component scores for each PCA factor were calculated with a regression analysis. The results are illustrated in Fig. 3. A two-way ANOVA analysis of the difference of emotions according to the type of park and the change of time demonstrated that the soundscape emotions had statistically significant differences only for the type of park. Only tranquil/peaceful had a significant interaction effect with park type and time change. Overall, tranquil/peaceful was about 0.34 in Park A, 0.63 and 0.39 higher than Park C and Park J, respectively. Dynamic/lively was –0.31 in Park A, which was 0.57 and 0.36 lower than Park C and J, respectively, and confusing/unpleasant was –0.2, which was 0.40 and 0.20 lower than Park C and J, respectively. Therefore, Park A can be seen as relatively calm compared to other parks, but where dynamic changes rarely happen.

In contrast, in Park C, many people were using the park, and there were many dynamic events at tourist spots; consequently, the dynamic/lively component was the highest at 0.26, and it was the lowest for tranquil/peaceful. Compared with other parks, the confusing/unpleasant component was relatively high. However, at 0.20, the result of the Likert scale response cannot be evaluated negatively. In other words, the participants' perception of the park was relatively positive. At 0.24, Park J had a higher tranquil/peaceful degree than Park C, but the remaining components had values between Park A and Park C, although the sound and visual indicators were similar to Park C. This was consistent with the findings of previous studies that the water sound

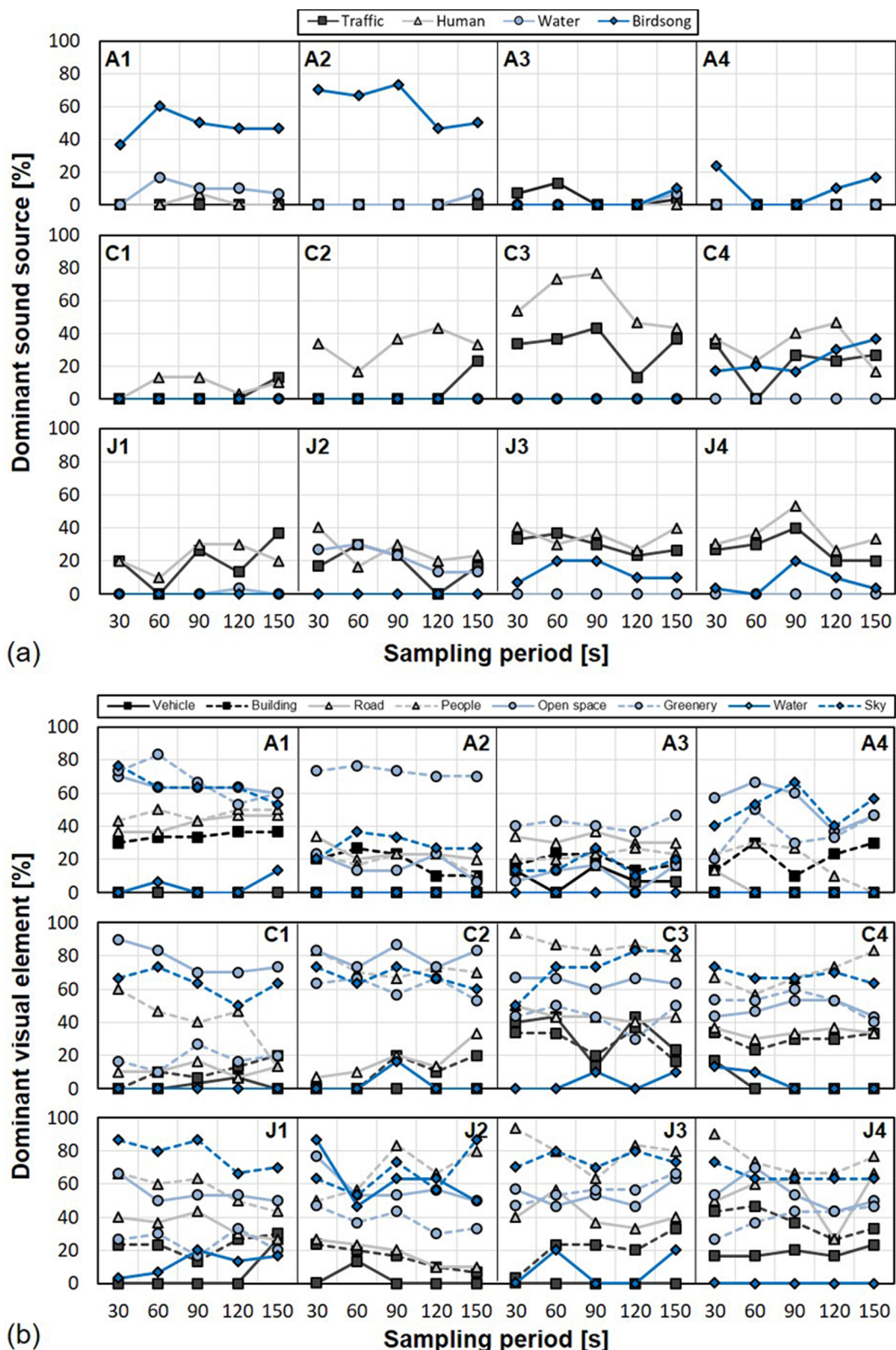


Fig. 2. Temporal variation in perceived audio and visual elements for different parks: (a) dominant sound sources, (b) dominant visual elements.

positively affects acoustic comfort and loudness perception (Jeon et al., 2010).

### 3.1.3. cognition—overall quality

Fig. 4 illustrates the results of the participants' cognition responses to the soundscape of the urban park. As a result of the two-way ANOVA

analysis on the cognition responses according to the type of park and time, the cognition was significantly different according to the type of park, but there was no statistically significant difference for time ( $p > 0.05$ ). In Park A, the perceived loudness and perceived crowd density were the lowest at  $-0.34$  and  $-0.39$ , respectively, and the comfort was highest at  $0.35$ . This demonstrates that the participants

**Table 3**  
Main and Interaction Effects of Location and Time for Different Audio and Visual Indicators.

| Indicator              | Factor      | SS     | df | MS     | F      | p      | ES( $\eta_p^2$ ) |
|------------------------|-------------|--------|----|--------|--------|--------|------------------|
| <i>Sound sources</i>   |             |        |    |        |        |        |                  |
| Traffic                | Park        | 279.26 | 2  | 139.63 | 135.69 | < 0.01 | 0.13             |
|                        | Time        | 12.67  | 4  | 3.17   | 3.08   | < 0.05 | 0.01             |
|                        | Interaction | 32.57  | 8  | 4.01   | 3.96   | < 0.01 | 0.02             |
| Human                  | Park        | 689.25 | 2  | 344.63 | 337.11 | < 0.01 | 0.27             |
|                        | Time        | 8.54   | 4  | 2.14   | 2.09   | 0.08   | 0.01             |
|                        | Interaction | 14.21  | 8  | 1.78   | 1.74   | 0.09   | 0.01             |
| Water                  | Park        | 59.61  | 2  | 29.81  | 55.88  | < 0.01 | 0.06             |
|                        | Time        | 2.14   | 4  | 0.54   | 1.00   | 0.40   | 0.01             |
|                        | Interaction | 6.10   | 8  | 0.76   | 1.43   | 0.18   | 0.01             |
| Birdsong               | Park        | 343.11 | 2  | 171.55 | 150.90 | < 0.01 | 0.15             |
|                        | Time        | 1.85   | 4  | 0.46   | 0.41   | 0.80   | 0.01             |
|                        | Interaction | 4.68   | 8  | 0.58   | 0.51   | 0.85   | 0.01             |
| <i>Visual elements</i> |             |        |    |        |        |        |                  |
| Vehicle                | Park        | 75.67  | 2  | 37.84  | 46.52  | < 0.01 | 0.05             |
|                        | Time        | 2.54   | 4  | 0.64   | 0.78   | 0.54   | 0.01             |
|                        | Interaction | 8.13   | 8  | 1.02   | 1.25   | 0.27   | 0.01             |
| Building               | Park        | 70.30  | 2  | 35.15  | 38.13  | < 0.01 | 0.04             |
|                        | Time        | 1.52   | 4  | 0.38   | 0.41   | 0.80   | 0.01             |
|                        | Interaction | 4.89   | 8  | 0.61   | 0.66   | 0.73   | 0.01             |
| Road                   | Park        | 57.59  | 2  | 28.79  | 34.62  | < 0.01 | 0.04             |
|                        | Time        | 1.20   | 4  | 0.30   | 0.36   | 0.84   | 0.01             |
|                        | Interaction | 4.36   | 8  | 0.55   | 0.66   | 0.73   | 0.01             |
| People                 | Park        | 175.63 | 2  | 87.82  | 171.96 | < 0.01 | 0.16             |
|                        | Time        | 2.02   | 4  | 0.51   | 0.99   | 0.41   | 0.01             |
|                        | Interaction | 3.38   | 8  | 0.42   | 0.83   | 0.58   | 0.01             |
| Open space             | Park        | 261.30 | 2  | 130.65 | 144.56 | < 0.01 | 0.13             |
|                        | Time        | 7.45   | 4  | 1.86   | 2.06   | 0.08   | 0.01             |
|                        | Interaction | 3.28   | 8  | 0.41   | 0.45   | 0.89   | 0.01             |
| Greenery               | Park        | 20.60  | 2  | 10.30  | 10.72  | < 0.01 | 0.01             |
|                        | Time        | 1.47   | 4  | 0.37   | 0.38   | 0.82   | 0.01             |
|                        | Interaction | 4.74   | 8  | 0.59   | 0.62   | 0.77   | 0.01             |
| Water                  | Park        | 232.34 | 2  | 116.17 | 134.37 | < 0.01 | 0.13             |
|                        | Time        | 2.21   | 4  | 0.55   | 0.64   | 0.64   | 0.01             |
|                        | Interaction | 9.40   | 8  | 1.18   | 1.36   | 0.21   | 0.01             |
| Sky                    | Park        | 130.29 | 2  | 65.15  | 75.89  | < 0.01 | 0.08             |
|                        | Time        | 2.89   | 4  | 0.72   | 0.84   | 0.50   | 0.01             |
|                        | Interaction | 2.90   | 8  | 0.36   | 0.42   | 0.91   | 0.01             |

**Table 4**  
A Rotated Component Matrix of PCA Using Semantic Expression in Terms of Soundscape (Numbers in Parentheses Represent Explained Variance).

| Attributes                  | Component 1<br>(26.30) | Component 2<br>(24.00) | Component 3<br>(18.48) |
|-----------------------------|------------------------|------------------------|------------------------|
| <i>Tranquil/peaceful</i>    |                        |                        |                        |
| Uneventful                  | 0.74                   | −0.28                  | 0.19                   |
| Monotonous                  | 0.82                   | −0.10                  | −0.07                  |
| Calm                        | 0.83                   | 0.08                   | −0.20                  |
| <i>Dynamic/lively</i>       |                        |                        |                        |
| Pleasant                    | 0.23                   | 0.75                   | 0.14                   |
| Eventful                    | −0.32                  | 0.78                   | 0.12                   |
| Exciting                    | −0.22                  | 0.78                   | 0.15                   |
| <i>Confusing/unpleasant</i> |                        |                        |                        |
| Unpleasant                  | 0.01                   | 0.12                   | 0.84                   |
| Chaotic                     | −0.10                  | 0.19                   | 0.80                   |

<sup>a</sup>Principal Component Analysis.

expressed the highest satisfaction and felt more comfortable with a quiet and peaceful park environment. Park C, however, had the highest perceived loudness and perceived crowd density at 0.27 and 0.15, respectively, due to the sounds from many people in the park. Comfort

was measured at 0.22, which was relatively low, but positive results were obtained on the Likert scale, as confirmed for the soundscape emotions. The participants perceived the place as comfortable. Finally, Park J had values between Park A and Park C with cognition scores of −0.01, 0.03, and 0.27, respectively. The participants perceived the park as having characteristics between a “calm park (A)” and an “active park (C).”

To summarize the results of the subjective evaluation of the parks, of all the audio and visual elements, those with a human source, that is, sounds made by people and the visual presence of people, were the most recognized. The overall quality of soundscapes in each park was significantly different from those of the other parks. Finally, each park evoked different soundscape emotions resulting from the people-generated audio and visual elements.

### 3.2. Human behavioral characteristics with different activity criteria

#### 3.2.1. Overall human behavioral characteristics according to social interaction and stay conditions

Various human behavioral characteristics related to social interaction and stay conditions in the parks are illustrated in Fig. 5. First, Park A had a minimal number of people compared to the other parks. From Fig. 5(a), the proportion of people who used the park alone was very high, and most people passed through without staying. Park C and Park J had a higher proportion of people who visited the park with other people, and there were groups of up to 25 (J3). Therefore, Park C and Park J are expected to have more social interactions. The degree of activity of this social interaction can be inferred from the fact that the percentages of stoppers staying in Park C and Park J in Fig. 5(b) are higher than those of the passers. As we moved from C1 to C4 in Park C, the total number of people increased, especially the number of groups. This may be because the Eiffel Tower is located in C4. The difference in human behavioral characteristics according to the type of park was statistically significant ( $p < 0.01$ ), but the difference in temporal variation was only statistically significant for the passers (see Table 5). The results demonstrated that there was a statistically significant difference in human behavioral characteristics according to time change only in Park C and Park J.

#### 3.2.2. Detailed human behavioral characteristics patterns according to the types of activity

Human behavioral characteristics in the park were classified into five categories according to the type of activity (Fig. 6). First, the most common behavior was to sit or rest and to stroll. There was very little human activity in Park A, and on average, there were very few people resting (5.55). In Park C, the number of people who stayed in place and walked in the park averaged 53.05 and 67.95, respectively, indicating no significant difference. In Park J, the average number of people stopping was 53.30, which is similar to Park C, but the number of walking people averaged 124.15, which was more than double that of Park C. Less than ten people exhibited other behaviors (loitering, jogging/running, and cycling), and these were only in Park C and Park J.

These results demonstrated that the behavior patterns of users changed according to the function and role of the park, and the difference in the behavior patterns in each park was statistically significant ( $p < 0.01$ ; see Table 5). In particular, walking, jogging/running, and cycling had statistically significant differences with time. However, there was an interaction effect on the type of park. The result of analyzing the simple main effect according to the time change by each park showed that there were significant differences in walking in Park C, jogging/running in Park C, jogging/running in Park J, and cycling in Park J. It was interesting to note that the characteristic feature of behaviors with other significant differences in time change was that the size (or radius) of the activity was higher than that of walking.



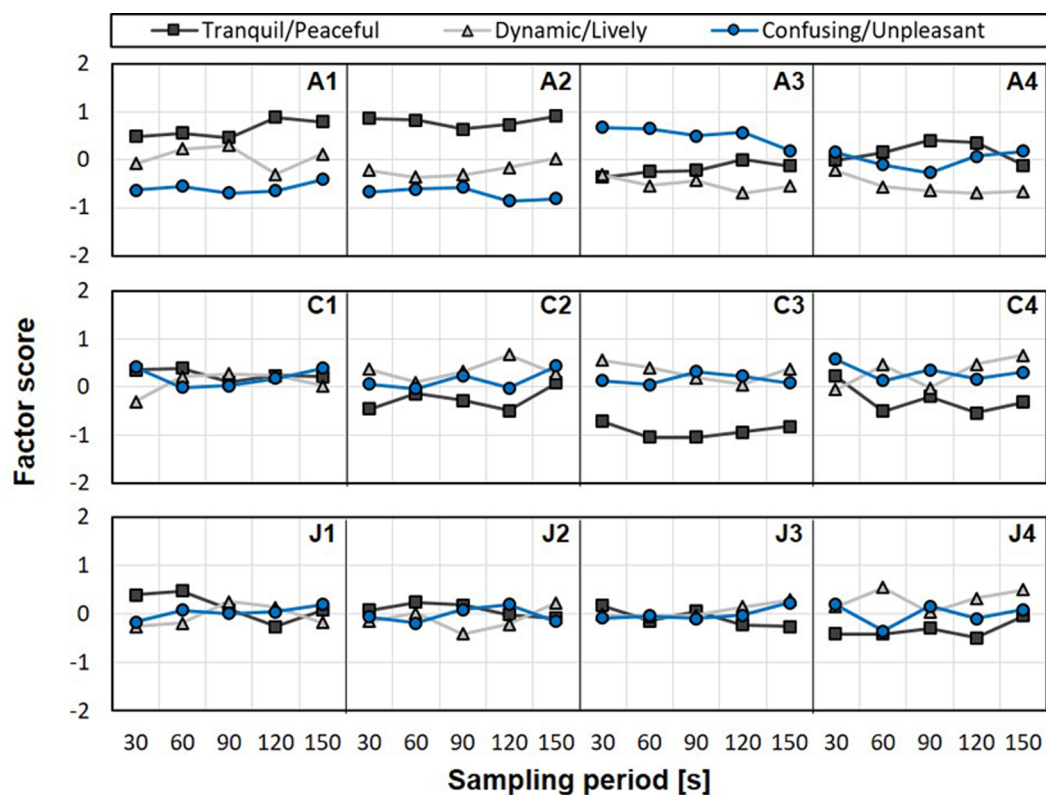


Fig. 3. Temporal variation of the mean principal component analysis scores for the three parks.

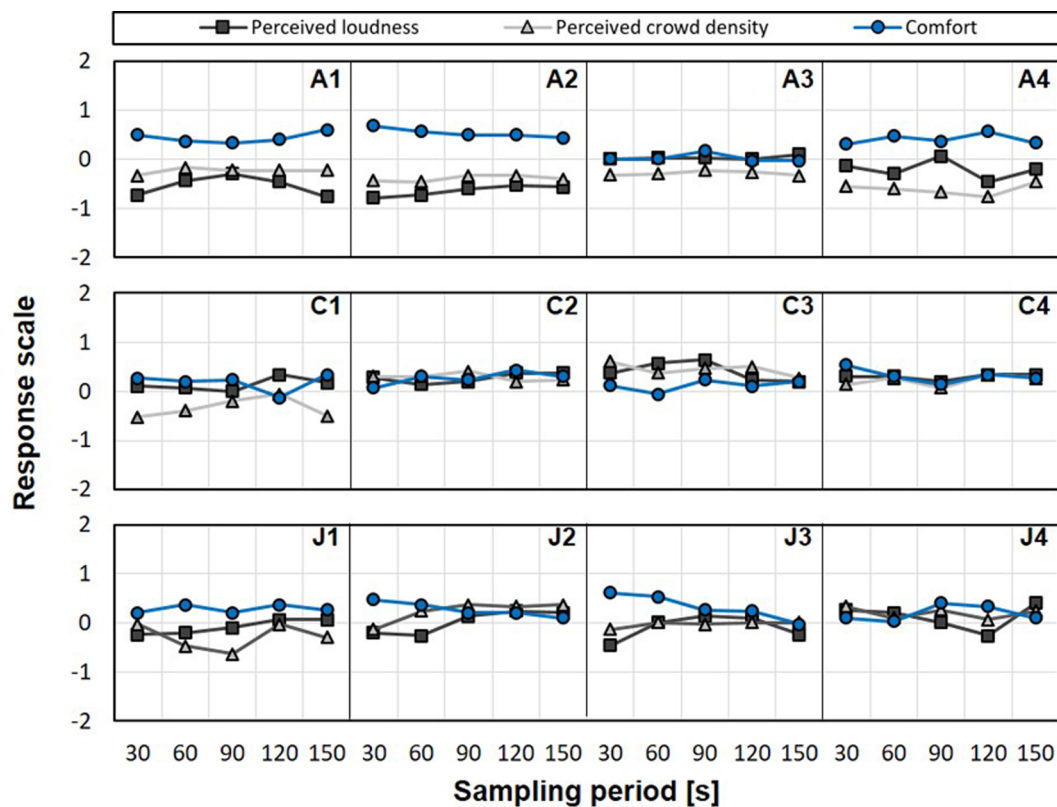
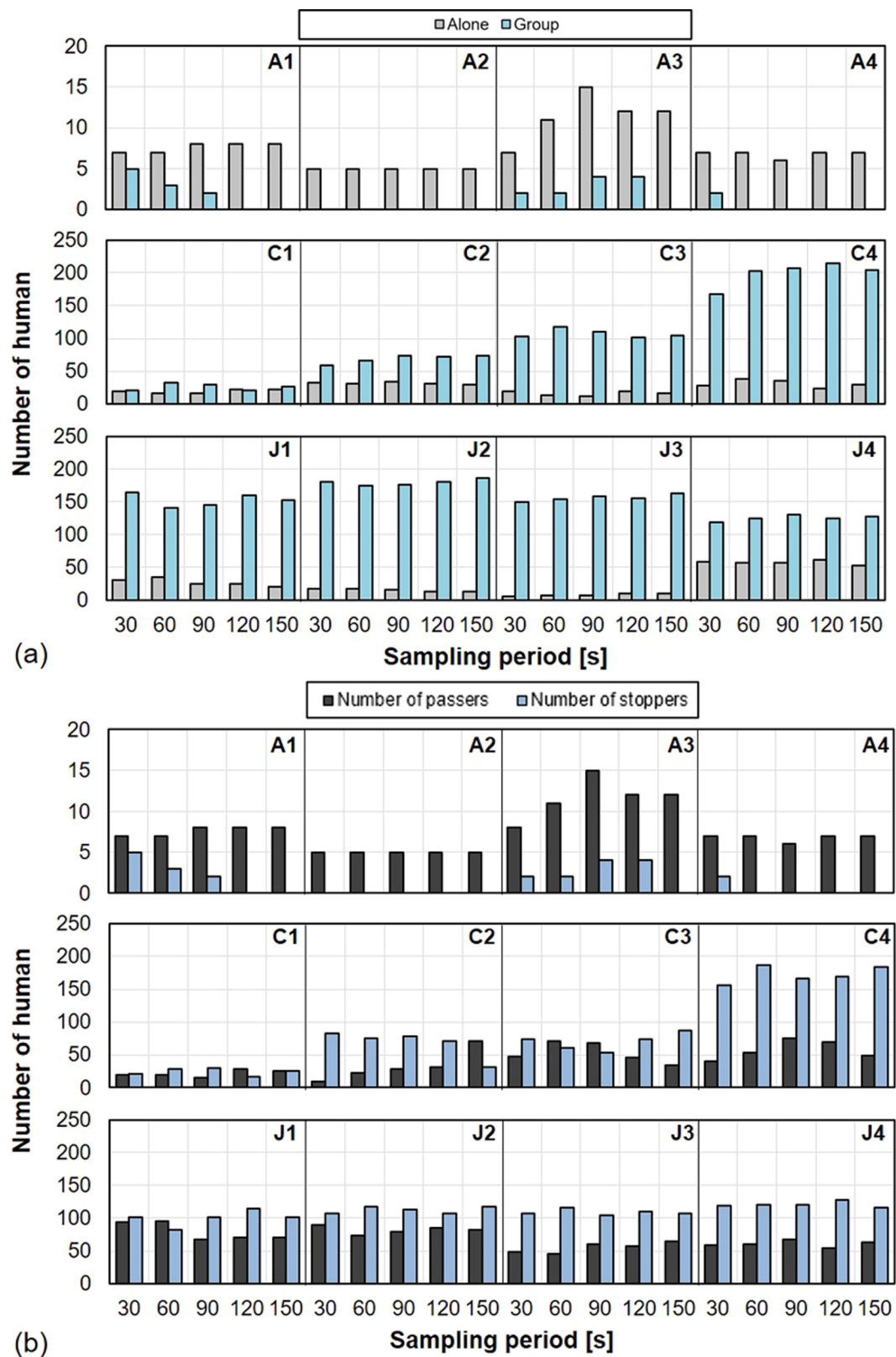


Fig. 4. Temporal variation of the mean response scale for the three parks.



**Fig. 5.** Temporal variation in the number of people for the three parks according to different criteria: (a) social interaction—alone or group; (b) stay conditions—passer or stopper.

**Table 5**  
Main and Interaction Effects of Location and Time for Different Soundscape Emotions.

| Indicator                 | Factor      | SS         | df | MS         | F       | p      | ES( $\eta_p^2$ ) |
|---------------------------|-------------|------------|----|------------|---------|--------|------------------|
| <i>Social interaction</i> |             |            |    |            |         |        |                  |
| <i>Alone</i>              | Park        | 134848.00  | 2  | 67424.00   | 477.62  | < 0.01 | 0.35             |
|                           | Time        | 465.00     | 4  | 116.25     | 0.82    | 0.51   | 0.01             |
|                           | Interaction | 1572.00    | 8  | 196.50     | 1.40    | 0.20   | 0.01             |
| <i>Group</i>              | Park        | 7152844.00 | 2  | 3576422.00 | 2352.57 | < 0.01 | 0.73             |
|                           | Time        | 7037.00    | 4  | 1759.25    | 1.16    | 0.35   | 0.01             |
|                           | Interaction | 22441.00   | 8  | 2805.13    | 1.85    | 0.07   | 0.01             |
| <i>Stay</i>               |             |            |    |            |         |        |                  |
| <i>Passer</i>             | Park        | 1155700.00 | 2  | 577850.00  | 2953.73 | < 0.01 | 0.77             |
|                           | Time        | 6055.00    | 4  | 1513.75    | 7.74    | < 0.01 | 0.02             |
|                           | Interaction | 19835.00   | 8  | 2479.38    | 12.67   | < 0.01 | 0.05             |
| <i>Stopper</i>            | Park        | 3904513.00 | 2  | 1952256.50 | 1811.14 | < 0.01 | 0.67             |
|                           | Time        | 1150.00    | 4  | 287.50     | 0.27    | 0.90   | 0.01             |
|                           | Interaction | 4907.00    | 8  | 613.38     | 0.57    | 0.80   | 0.01             |
| <i>Activity</i>           |             |            |    |            |         |        |                  |
| <i>Stopping</i>           | Park        | 907275.00  | 2  | 453637.50  | 1095.88 | < 0.01 | 0.55             |
|                           | Time        | 1313.00    | 4  | 328.25     | 0.79    | 0.53   | 0.01             |
|                           | Interaction | 4330.00    | 8  | 541.25     | 1.31    | 0.24   | 0.01             |
| <i>Loitering</i>          | Park        | 3969.00    | 2  | 1984.50    | 824.28  | < 0.01 | 0.48             |
|                           | Time        | 33.00      | 4  | 8.25       | 3.43    | 0.01   | 0.01             |
|                           | Interaction | 66.00      | 8  | 8.25       | 3.43    | 0.01   | 0.02             |
| <i>Walking</i>            | Park        | 4470661.00 | 2  | 2235330.50 | 3890.77 | < 0.01 | 0.81             |
|                           | Time        | 13887.00   | 4  | 3471.75    | 6.04    | < 0.01 | 0.01             |
|                           | Interaction | 25674.00   | 8  | 3209.25    | 5.59    | < 0.01 | 0.02             |
| <i>Jogging/running</i>    | Park        | 687.00     | 2  | 343.50     | 209.09  | < 0.01 | 0.19             |
|                           | Time        | 38.00      | 4  | 9.50       | 5.78    | < 0.01 | 0.01             |
|                           | Interaction | 148.00     | 8  | 18.50      | 11.26   | < 0.01 | 0.05             |
| <i>Cycling</i>            | Park        | 379.00     | 2  | 189.50     | 74.58   | < 0.01 | 0.08             |
|                           | Time        | 457.00     | 4  | 114.25     | 44.95   | < 0.01 | 0.09             |
|                           | Interaction | 716.00     | 8  | 89.50      | 35.21   | < 0.01 | 0.14             |

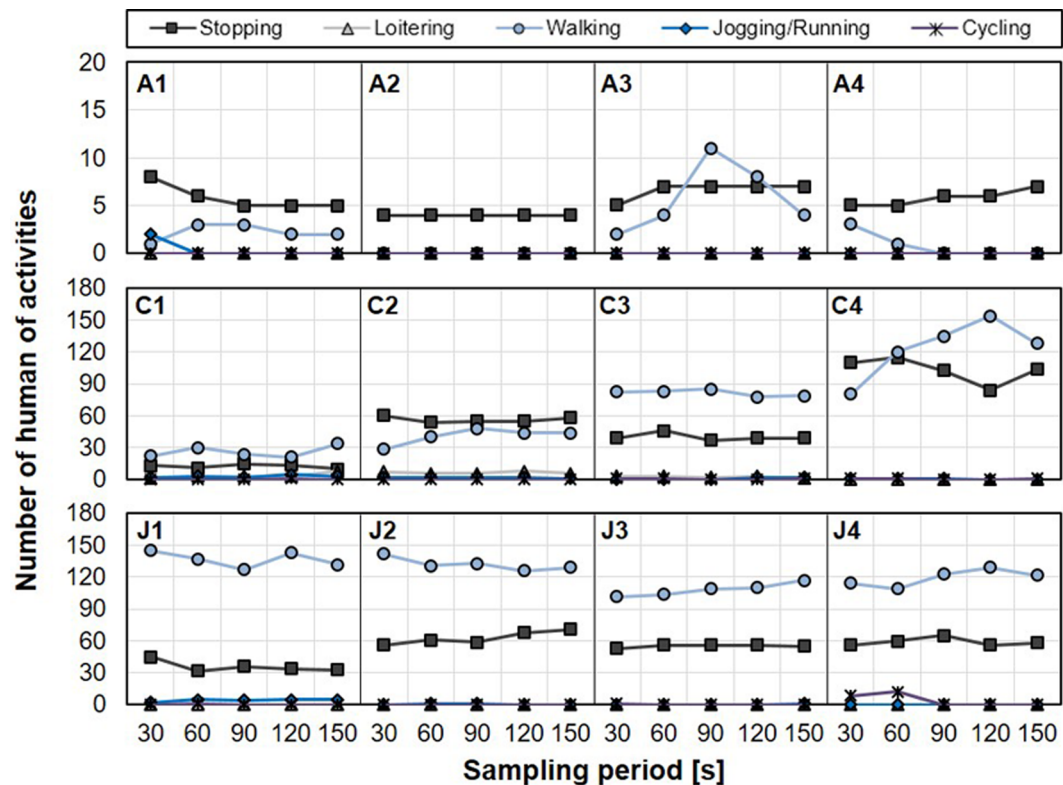


Fig. 6. Temporal variation of human behavior patterns for the three parks.



**Table 6**  
Pearson Correlation Coefficient between Audio and Visual Indicator and Soundscape Quality (Correlation Value in Parentheses Based on the Time Variation of Each Parameter).

|                         | Emotion           |            |                |            |                      | Cognition          |       |                         |       |            |
|-------------------------|-------------------|------------|----------------|------------|----------------------|--------------------|-------|-------------------------|-------|------------|
|                         | Tranquil/peaceful |            | Dynamic/lively |            | Confusing/unpleasant | Perceived loudness |       | Perceived crowd density |       | Comfort    |
| <b>Sound source</b>     |                   |            |                |            |                      |                    |       |                         |       |            |
| Traffic ( $\Delta$ )    | -0.57             | ** (0.22)  | 0.41           | ** (-0.16) | 0.39                 | ** (0.27)          | 0.53  | ** (0.06)               | 0.55  | ** (-0.23) |
| Human ( $\Delta$ )      | -0.69             | ** (-0.16) | 0.68           | ** (0.05)  | 0.30                 | *                  | 0.67  | ** (0.22)               | 0.79  | ** (0.03)  |
| Birdsong ( $\Delta$ )   | 0.56              | ** (-0.18) | -0.02          | ** (0.19)  | -0.71                | ** (0.38)          | -0.65 | ** (0.16)               | -0.25 | ** (0.04)  |
| Water ( $\Delta$ )      | 0.29              | * (-0.01)  | -0.08          | (0.05)     | -0.31                | *                  | -0.24 | (0.03)                  | 0.07  | (0.13)     |
| <b>Visual elements</b>  |                   |            |                |            |                      |                    |       |                         |       |            |
| Vehicle ( $\Delta$ )    | -0.64             | ** (0.10)  | 0.29           | * (-0.01)  | 0.40                 | ** (0.15)          | 0.51  | ** (-0.09)              | 0.46  | ** (0.21)  |
| Building ( $\Delta$ )   | 0.07              | (-0.01)    | -0.02          | (-0.15)    | -0.24                | (0.19)             | -0.27 | * (-0.16)               | -0.03 | (0.15)     |
| Road ( $\Delta$ )       | -0.20             | (0.07)     | 0.36           | ** (0.24)  | -0.10                | (0.16)             | 0.01  | (0.29)                  | *     | (0.23)     |
| People ( $\Delta$ )     | -0.50             | ** (-0.02) | 0.72           | ** (0.04)  | 0.10                 | (-0.09)            | 0.48  | ** (0.26)               | 0.79  | ** (0.22)  |
| Open space ( $\Delta$ ) | -0.20             | (-0.03)    | 0.48           | ** (0.08)  | 0.07                 | (-0.21)            | 0.35  | ** (0.01)               | 0.32  | * (-0.05)  |
| Greenery ( $\Delta$ )   | 0.18              | (-0.10)    | 0.18           | (-0.07)    | -0.47                | ** (0.08)          | -0.37 | ** (-0.41)              | 0.21  | ** (-0.03) |
| Water ( $\Delta$ )      | -0.01             | (-0.16)    | 0.03           | (-0.04)    | 0.02                 | (0.26)             | 0.07  | (0.14)                  | 0.32  | *          |
| Sky ( $\Delta$ )        | -0.18             | (-0.15)    | 0.50           | ** (-0.09) | -0.07                | (-0.07)            | 0.26  | *                       | 0.36  | ** (-0.06) |

## 4. Discussion

### 4.1. The effect of audio and visual indicators on soundscape perception

Table 6 illustrates the correlation between the sound source and visual elements and soundscape emotions and cognition. For a more meaningful interpretation, we examined the relationship that demonstrates the value of the correlation coefficient above the small effect (0.2) of Cohen's convention of a Pearson correlation (Cohen, West, & Aiken, 2014). First, it was found to have a positive effect on the evaluation of urban parks ( $p < 0.01$ ); with a value of 0.56, the sound source indicator demonstrated that birdsong has a strong positive correlation with tranquil/peaceful and comfort, and a strong negative correlation with confusing/unpleasant and perceived loudness, with values of  $-0.71$  and  $-0.65$ , respectively. Similar to birdsong, the water sound also had a positive effect on the evaluation of a park's calmness ( $\gamma = 0.29, p < 0.05$ ), indicating that natural sounds made a positive contribution to the soundscape evaluation of urban parks (De Coensel, Vanwetswinkel, & Botteldooren, 2011). For human sound, there was a strong positive relationship of 0.68 with the dynamic/liveliness of the park and a very high positive correlation of 0.79 with the perceived crowd density. Contrary to natural sound, the human sound had a negative relationship with a park's tranquil/peaceful ( $\gamma = -0.69, p < 0.01$ ) and comfort ( $\gamma = -0.28, p < 0.05$ ) levels, and a positive relationship with perceived loudness ( $\gamma = 0.67, p < 0.01$ ) and confusing/unpleasant ( $\gamma = 0.30, p < 0.05$ ). Therefore, it was confirmed that human sound plays an important role in determining emotions and cognition related to the calmness or liveliness of park soundscapes (Axelsson, 2015; Axelsson, Nilsson, Hellström, & Lundén, 2014).

Looking at the relationship between sound sources and visual elements, traffic noise had a high positive correlation with vehicles ( $\gamma = 0.65, p < 0.01$ ), people ( $\gamma = 0.60, p < 0.01$ ), and roads ( $\gamma = 0.56, p < 0.01$ ). Human sound also had a high correlation with people ( $\gamma = 0.80, p < 0.01$ ), vehicles ( $\gamma = 0.50, p < 0.01$ ), and roads ( $\gamma = 0.41, p < 0.01$ ), from which it is possible to determine that human activities that caused sound occurred near the roads. Among the natural sounds, birdsongs had a positive correlation with greenery ( $\gamma = 0.68, p < 0.01$ ) and a negative correlation with vehicles ( $\gamma = -0.43, p < 0.01$ ), while water sounds only demonstrated a high positive correlation with water spaces ( $\gamma = 0.75, p < 0.01$ ). One interesting fact was found in the relationship between changes in the audio and visual indicator identification rate and changes in emotions and cognition over time. The change in birdsong was positively related to the amount of confusing/unpleasant change. Although birdsong can be used to influence the complex and uncomfortable sound environment of a park, randomly increasing birdsong over time can have a negative impact on a park's sound environment.

Next, for visual elements, vehicles tended to be similar to traffic sounds, people were similar to humans, and birdsong was similar to greenery. Many visual indicators had a positive correlation with dynamic/lively, and a visual identification of people was the most critical factor ( $\gamma = 0.72, p < 0.01$ ). Roads, open spaces, and sky also had positive correlations with values of 0.36, 0.48, and 0.50, respectively. This can be explained by the fact that these represented places where people's activities occur in the park. In the case of the sky, the value increased with the increase in open spaces.

Similarly, we found that a visual identification of people had a strong positive correlation with perceived loudness and perceived crowd density. Overall, it was found that audio and visual identification of people is a very important factor in assessing the soundscape of the park (Meng & Kang, 2015). The relationship between the rate of change of visual identification and emotions and cognition for time change was as follows. When the identification rate for greenery increased, participants perceived less sound loudness psychologically ( $\gamma = -0.41, p < 0.01$ ), and comfort also increased ( $\gamma = 0.43, p < 0.01$ ). This was

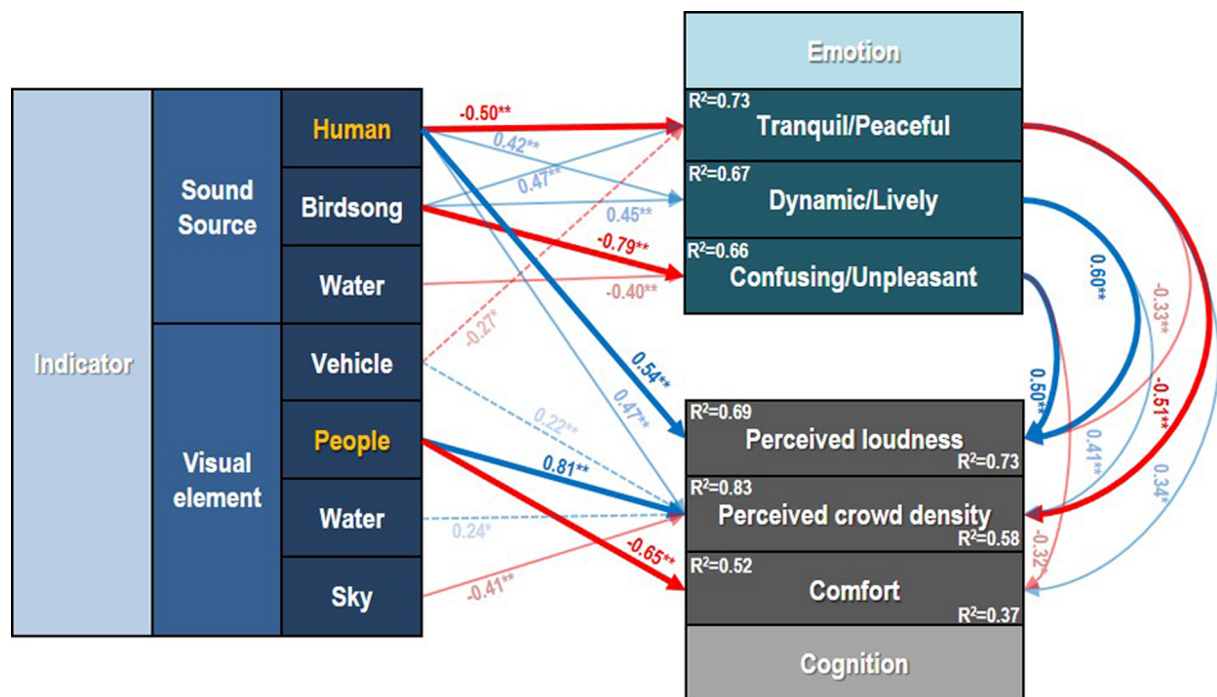


Fig. 7. Soundscape perceptual model in urban parks (red color: negative contribution; blue color: positive contribution; \* $p < 0.05$ , \*\* $p < 0.01$ ). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

consistent with the results of the positive effects that the sound of birdsong had on park soundscapes (Van Renterghem, 2018).

#### 4.2. Soundscape perception model in urban parks

As illustrated in Fig. 7, a multiple linear regression (MLR) analysis was performed to examine the relationships and contributions among the indicators (four sound sources and eight visual elements), three emotions, and three cognitions on soundscape, and the audio (traffic noise) and visual indicators (buildings, roads, open spaces, and greenery) demonstrating that the statistically insignificant regression coefficients were excluded. We conducted the MLR analysis for three cases: 1) A regression model with the indicators as the independent variable and emotions as the dependent variable; 2) a regression model with the indicators as the independent variable and cognition as the dependent variable; and 3) a regression model with emotions as the independent variable and cognition as the dependent variable. The overall model was statistically significant ( $p < 0.01$ ), and variables had an  $R^2$  of 0.5 or more, indicating that more than 50% of the variation of the dependent variable could be predicted or explained by the independent variables. Overall, human sound, birdsong, and the presence of people had the greatest contribution to soundscape emotions and cognition.

First, among the audio and visual indicators that make up the context of the park, sound factors have a greater effect on soundscape emotions than visual factors. Tranquil/peaceful levels made statistically significant contributions to human sound ( $\beta = -0.50$ ,  $p < 0.01$ ), birdsong ( $\beta = 0.47$ ,  $p < 0.01$ ), and vehicle sound ( $\beta = -0.27$ ,  $p < 0.05$ ). This confirmed once again that human sound had a significant effect on the park's calmness. Additionally, human sound and birdsong contributed positively to dynamic/lively levels at 0.42 and 0.47, respectively. This was somewhat consistent with the results of previous studies (Aletta & Kang, 2018) that demonstrated people as the main factor determining vibrancy in evaluating urban soundscapes. However, the positive relationship between human sound and confusing/unpleasant emotions, as illustrated in Table 7, was not statistically significant. This suggests that human sound is relatively less

conductive to causing negative emotions such as park discomfort. This can be inferred from Davies et al. (2013) in the fact that human speech or happy human sound creates a positive emotional state. Natural sounds (birdsong:  $\beta = -0.79$ ,  $p < 0.01$ ; water:  $\beta = -0.40$ ,  $p < 0.01$ ) have been demonstrated to be effective in mitigating the negative emotions.

The soundscape perception models found interesting results. Unlike emotions, visual indicators are influenced a lot, and among sound source indicators, only human sound ( $\beta = 0.54$ ,  $p < 0.01$ ) had a significant contribution to perceived loudness and perceived crowd density at 0.47 and 0.81, respectively. With a value of  $-0.65$ , comfort had a negative effect on the visual identification of people. However, our result was consistent with the results (Meng & Kang, 2016) that corresponded to a decrease in comfort when the density of people increased to more than 0.25 persons/m<sup>2</sup> within a shopping street.

Finally, the predictive model between emotions and cognition demonstrates the park's calmness had negative contributions to loudness ( $\beta = -0.33$ ,  $p < 0.01$ ) or crowd density ( $\beta = -0.51$ ,  $p < 0.01$ ), while demonstrating positive contributions to comfort ( $\beta = 0.34$ ,  $p < 0.01$ ). The dynamics of the park contributed 0.60 and 0.41 to loudness and crowd density, respectively. Confusing/unpleasant also contributed 0.50 ( $p < 0.01$ ) and  $-0.32$  ( $p < 0.05$ ) to perceived loudness and comfort, respectively.

Taken together, it is clear that the audio and visual perception of a person plays a very important role in the emotion and cognition of soundscape. The reason is that due to the spatial nature of the park (except for vehicles), people move with the highest levels of activity when compared to other elements, and humans can relate to other people within a specified space through social activities. Therefore, even in the same space, the influence of the human being and the sound generated by the human being in recognizing the space is very strong (Jeon & Hong, 2015).

#### 4.3. Relationship between the temporal variation of human behavioral characteristics and soundscape perception

Table 7 illustrates the relationship between human behavioral

**Table 7**  
Pearson Correlation Coefficient between Human Behavior and Soundscape Quality (Correlation Value in Parentheses Based on the Time Variation of Each Parameter) (\* $p < 0.05$ , \*\* $p < 0.01$ ).

|                              | Cognition         |                   |                      |                    |                         |                    |
|------------------------------|-------------------|-------------------|----------------------|--------------------|-------------------------|--------------------|
|                              | Emotion           |                   | Perceived loudness   |                    | Perceived crowd density |                    |
|                              | Tranquil/peaceful | Dynamic/lively    | Confusing/unpleasant | Perceived loudness | Perceived crowd density | Comfort            |
| <i>Social interaction</i>    |                   |                   |                      |                    |                         |                    |
| Alone ( $\Delta$ )           | -0.39<br>**       | (-0.09)<br>**     | 0.48<br>**           | (-0.04)<br>**      | 0.28<br>*               | (-0.02)<br>*       |
| Group ( $\Delta$ )           | -0.44<br>**       | (0.42)<br>**      | 0.45<br>**           | (0.19)<br>**       | 0.28<br>*               | (-0.25)<br>*       |
| <i>Stay conditions</i>       |                   |                   |                      |                    |                         |                    |
| Passer ( $\Delta$ )          | -0.39<br>**       | (0.21)<br>**      | 0.35<br>**           | (-0.35)<br>**      | 0.28<br>*               | (0.27)<br>**       |
| Stopper ( $\Delta$ )         | -0.48<br>**       | (-0.53)<br>**     | 0.53<br>**           | (0.46)<br>**       | 0.29<br>**              | (-0.44)<br>**      |
| <i>Activity</i>              |                   |                   |                      |                    |                         |                    |
| Stopping ( $\Delta$ )        | -0.48<br>**       | (0.01)<br>**      | 0.54<br>**           | (0.07)<br>**       | 0.33<br>**              | (-0.02)<br>**      |
| Loitering ( $\Delta$ )       | -0.26<br>*        | (-0.18)<br>**     | 0.39<br>**           | (0.08)<br>**       | 0.22<br>**              | (-0.12)<br>**      |
| Walking ( $\Delta$ )         | -0.43<br>**       | (-0.37)<br>**     | 0.43<br>**           | (0.03)<br>**       | 0.25<br>**              | (-0.12)<br>**      |
| Jogging/running ( $\Delta$ ) | 0.03<br>(0.11)    | (0.11)<br>(-0.05) | 0.17<br>(-0.02)      | (-0.05)<br>(0.36)  | 0.18<br>*               | (-0.02)<br>*       |
| Cycling ( $\Delta$ )         | -0.21<br>(-0.01)  | (-0.01)<br>(0.36) | 0.25<br>(0.36)       | (0.36)<br>(0.36)   | -0.03<br>(-0.41)        | (-0.41)<br>(-0.41) |

characteristics and soundscape perception. First, the relationship between soundscape emotions and behavior had a negative relationship with a park's calmness, a positive relationship with liveliness, and a negative relationship with confusion. In particular, the higher the number of people using the park, regardless of social interaction or stay conditions, the higher the eventfulness. However, for stay conditions, the relationship between dynamic/lively ( $\gamma = 0.53, p < 0.05$ ) and the number of people staying in the park was stronger than the relationship with the number of passers ( $\gamma = 0.35, p < 0.05$ ). To increase eventfulness, parks should be designed to allow people to stay. These new findings also become apparent when we examine the relationship between emotions and cognition based on the type of activity. In general, the more activities (stopping, loitering, and walking) there were with a smaller radius of movement than activities (jogging/running and cycling) with a bigger radius, the more spaces were experienced as eventful.

Interestingly, the amount of change in the pattern of human behavioral characteristics over time was also evident (see parentheses in Table 7). Park calmness ( $\gamma = -0.53, p < 0.05$ ) decreased with an increasing number of stoppers over time, but vitality ( $\gamma = 0.46, p < 0.05$ ) increased along with discomfort ( $\gamma = -0.44, p < 0.05$ ). The decrease was found to have a positive effect on people's perception of soundscapes. In contrast, as the number of passers increased with time, the liveliness of the park decreased ( $\gamma = 0.46, p < 0.05$ ).

Regarding the relationship between soundscape cognition and human behavior, there was a significant positive correlation with perceived loudness and crowd density. However, there was no clear relationship with comfort except for the number of people visiting the park alone ( $\gamma = -0.26, p < 0.05$ ). As illustrated in the soundscape perceptual model in Fig. 7, this trend did not appear in the classification of activity, although it was clear that the presence of people contributes to comfort. It can be inferred that the important factor in determining the preference of the urban park soundscape is the presence or absence of people and not specific behaviors. Nevertheless, as discussed above, it is evident that the social behavior of interaction, stay, and activity type can influence soundscape perception, with the exception of comfort.

#### 4.4. Limitations and further research

The behavioral impacts were evaluated for a relatively short time and failed to sufficiently reflect the changes in biophony, depending on morning and afternoon hours. Additionally, it is necessary to further examine the effects of various contexts other than those examined in this study, such as the individual's socio-cultural background or the frequency of park visits (Liu et al., 2013). Although this study has quantified changes in human behavioral characteristics and grouped them in various standards through observation, it is limited in that it did not differentiate between the sounds caused by human behavioral characteristics in the auditory sensory domain. There may be discoveries if human sounds are distinguished between speech-related sounds (verbal sounds) and activity-related sounds (non-verbal sounds; Aletta, Kang, Astolfi, & Fuda, 2016; Aletta, Kang, Fuda, & Astolfi, 2016). Furthermore, it would be possible to analyze other human-related descriptors, such as safety and social presence besides cognition, which was considered in this study (Calleri et al., 2019; Sayin, Krishna, Ardelet, Briand Decré, & Goudey, 2015). Despite these limitations, this study is meaningful in that it discovered that the presence and behavioral changes of people affect the soundscape perception of urban parks. As the relationship between soundscape and human behavioral characteristics was considered a challenge in soundscape research, this study contributes to the expansion of soundscape studies related to human behavioral characteristics in a broad sense and has academic significance as an attempt at comprehensive research (Aletta & Xiao, 2018).



## 5. Conclusions

The purpose of this study was to investigate the effect of human behavioral characteristics on soundscape perception in urban parks. The study evaluated soundscape indicators, emotions, and cognition in urban parks in Paris, and examined their relationship with human behavioral characteristics analyzed through observations. A subjective evaluation of the three parks with different functions examined audio and visual elements (indicators), semantic expression (emotions), and soundscape preference (cognition) as perceived by participants. Observations were used to classify human behaviors based on social interactions, stay conditions, and activity types. The relationship between subjective evaluation and the importance of human behavioral characteristics for urban park soundscapes were newly investigated.

As a result of the subjective evaluation of urban park soundscapes, the identification of birdsong and human sounds were high as sound indicators, and only human sound was recognized differently in time. For the visual indicators, the participants focused on greenery and the presence of people. The semantic expression of the soundscape emotion included tranquil/peaceful, dynamic/lively, and confusing/unpleasant evaluations. Birdsong and human sounds contributed to the park's calmness and vitality, and natural sounds (birdsong and water) were found to be used positively to reduce confusing/unpleasant feelings. Furthermore, through the results of the soundscape cognition, we discovered that the most significant determinants of perceived loudness, crowd density, and comfort in a park are sounds made by people and the presence of people.

As a result of examining the human behavioral characteristics in the park, it was found that the higher the number of people in the park, the higher the proportion of people forming groups, resulting in more active social interactions. The results also demonstrated that staying in the park increased the proportion of people who participated in activities. The resulting increase in human activity had a negative relationship with calmness, a positive relationship with liveliness, and a negative relationship with confusion, with regards to the park's soundscape emotions. Additionally, for cognition, there was a strong positive relationship between perceived loudness and perceived crowd density.

The new findings relating to human behavioral influence on the perception of soundscapes in urban parks were as follows: 1) The greater the number of people who used the parks, regardless of whether they stayed or not, and the more their activities did not involve great movement (walking or less movement), the higher the perception of vibrancy in the park; 2) as the number of stoppers increased over time, calmness tended to decrease, vibrancy increased, and discomfort tended to decrease; as the number of passers increased, the liveliness decreased; and 3) the most important factor that determines the comfort (preference) of the park soundscape is the presence of people itself. Specifically, the results demonstrated that the kind of activities taking place in the park does not significantly affect comfort. The findings of this study are expected to be used in the effective design and planning of urban parks on a micro-scale basis as new considerations of human behavioral characteristics, park function, and landscape layout.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix I

### Audio-visual recording

To implement a VR environment for evaluating the soundscapes of urban parks in a laboratory environment, audio and visual information was collected from the 12 evaluation sites from 10 am to 2 pm in early October 2019. A spherical panoramic camera (INSTA 360, China) was used to collect three-dimensional visual information, for 150 s at 360 degrees, using 8 k ultra-high-definition, 80 fps resolution, and a bit-rate of 95 Mbps. An ambisonic microphone (Soundfield SPS 200) and a portable sound recorder (Mixpre-6, Sounddevices) were used to collect audio information, using A-format first-order ambisonic (FOA) simultaneously (Maffei et al., 2015; Sudarsono, Lam, & Davies, 2016). To calibrate the sound pressure level to be the same as the real environment when implementing the VR environment in the laboratory environment, the A-weight sound pressure level ( $L_{Aeq}$ ) was measured using a ½-inch microphone (G.R.A.S. AE46) and an AS-70 portable sound-level meter (RION). The average eye height of the 360-degree camera and the height of the ambisonic microphone, and the ½-inch microphone was 1.5 m. A clapper was used as a sound signal for synchronizing the visual information and the auditory information measured by the various pieces of equipment.

The physical acoustic environment characteristics measured at each evaluation point were analyzed and summarized according to three aspects and presented together in Table 1: sound strength, spectral contents, and the temporal structure of the sounds (De Coensel & Botteldooren, 2006). The sound source used in the analysis was the sound source measured by the correction microphone at each evaluation point. The A-weighted sound pressure level ( $L_{Aeq, 150\text{-sec}}$ ) was analyzed to determine the sound strength of the park sound environment.  $L_{Aeq, 150\text{-sec}}$  was analyzed to determine the relatively low-frequency characteristics related to sound spectral contents.  $L_{A10-90, 150\text{-sec}}$  was calculated to investigate the temporal variability of the sound environment. The results demonstrated that the sound pressure level was higher in the relatively crowded parks (C, J), but  $L_{Aeq, 150\text{-sec}}$  was lower in Park A, which was relatively tranquil. Additionally, the difference between the maximum and minimum values of  $L_{A10-A90}$  and  $L_{Ceq-Aeq}$  exceeded 10 dB, respectively, to sufficiently cover the range of variations of the acoustic environment that can occur in a general urban park.

### Reproduction method

To provide visual information for each of the 12 points, images recorded from 6 channels were made into a single 360-degree image through a post-process (using software Insta360stitcher) and linked to a head-mounted display (HMD). For the sound information, Google's Spatial Audio API built in Unity engine software version 2018.4 was used for the headphone based stereo sound. The A-format FOA recorded sound source was converted to B-format FOA and down-mixed to a binaural track. Next, the head tracker embedded in the HMD device was used to realize the real-time direction according to the subject's head

rotation to enhance the reality of the VR environment. The syncing of the images and sound was based on the clapper sound. To correct the sound pressure level at each evaluation point, 12 stereo sound sources were reproduced through headphones in a semi-anechoic chamber and recorded with a Brüel & Kjær 4100 head and torso simulator. Additionally, the A-weighted equivalent continuous sound level ( $L_{Aeq,150\text{-sec}}$ ) of each sound source was calculated. The sound pressure level was adjusted using Adobe Audition version 1.5 to match the  $L_{Aeq,150\text{-sec}}$  of the sound source recorded with the sound pressure correction microphone at the actual measurement site (Hong et al., 2019; Jo & Jeon, 2019; Sanchez et al., 2017).

## Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.landurbplan.2020.103890>.

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